F18CE2 - MV Calculus and Real Analysis B: Solutions 3

1 It is clear that 2 is a lower bound for A, as A only contains positive numbers, therefore $x^2 > 4$ implies $x > 2$. Let $a > 2$ be another lower bound for A. Then, we consider the point $x = \frac{2+a}{2}$, which is the middle point between 2 and a. We show that the point x belongs to A. It is clearly positive (in fact, $x > 2$). Since $a > 2$, we have that $x^2 = \frac{4+2a+a^2}{4} > \frac{16}{4} = 4$. This shows that there cannot be a lower bound for A that is bigger than 2, therefore 2 is the infimum.

2 This result can be proved by contradiction or through a direct proof. Here, we use a direct proof. Let L and M be two limit points of the sequence $\{a_n\}_{n\in\mathbb{N}}$. By the definition of limit, for any $\epsilon > 0$ there exists N_1 such that $|a_n - L| < \epsilon$ for any $n \geq N_1$, and there exists N_2 such that $|a_n - M| < \epsilon$ for any $n \geq N_2$. Therefore, for any $n \geq N = \max\{N_1, N_2\}$, we have both $|a_n - L| < \epsilon$ and $|a_n - M| < \epsilon$. Hence, for any $n \geq N$, we can write

$$|L = M| \leq |a_n - L| + |a_n - M| < 2\epsilon.$$

As we can do this for any $\epsilon > 0$, this implies that $|L - M| = 0$, i.e. $L = M$.

3 The sequence $\dfrac{(-1)^n}{n}$, $n = 1, 2, \ldots$, is convergent but not monotone.

The sequence (n) is increasing and not convergent.

4 We have that

$$b_{n+1} - b_n = (a_{n+2} - a_{n+1}) - (a_{n+1} - a_n) = a_{n+2} - 2a_{n+1} + a_n \geqslant 0.$$

Hence $b_{n+1} \geqslant b_n$ so that (b_n) is increasing.

5 An increasing sequence is convergent if it is bounded above.

A decreasing sequence is convergent if it is bounded below.

6 For all $n \geqslant 1$ we have that

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}\, n!}{2^n\, (n+1)!} = \frac{2}{n+1} \leqslant 1.$$

Hence $a_n \geqslant a_{n+1}$ so that (a_n) is decreasing.

Since a_n is positive, the sequence (a_n) is bounded below by 0 and hence converges.

7 For all $n \geqslant 1$ we have that

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^3\, 8^n}{n^3\, 8^{n+1}} = \frac{(n+1)^3}{8n^3} \leqslant \frac{(n+n)^3}{8n^3} = 1.$$

Hence $a_n \geqslant a_{n+1}$ so that (a_n) is decreasing.

Since a_n is positive, the sequence (a_n) is bounded below by 0 and hence converges.

8 Since $x_1 \geqslant 0$ and (x_n) increasing we have $x_n \geqslant 0$ for all $n \geqslant 1$, and then

$$\begin{aligned} a_{n+1} - a_n &= \frac{2x_{n+1}+3}{x_{n+1}+2} - \frac{2x_n+3}{x_n+2} \\ &= \frac{(2x_{n+1}+3)(x_n+2)) - (x_{n+1}+2)(2x_n+3)}{(x_{n+1}+2)(x_n+2)} \\ &= \frac{2x_{n+1}x_n + 4x_{n+1} + 3x_n + 6 - 2x_{n+1}x_n - 3x_{n+1} - 4x_n - 6}{(x_{n+1}+2)(x_n+2)} \\ &= \frac{x_{n+1} - x_n}{(x_{n+1}+2)(x_n+2)} \\ &\geqslant 0 \quad \text{(since } (x_n) \text{ is increasing)}, \end{aligned}$$

so (a_n) is increasing. Now,

$$2 - a_n = 2 - \frac{2x_n+3}{x_n+2} = \frac{2x_n+4-2x_n-3}{x_n+2} = \frac{1}{x_n+2} \geqslant 0,$$

so that $a_n \leqslant 2$ for all n. Thus (a_n) is increasing and bounded above and hence convergent.